Soil Oxygen Sensor introduction

SOR-01

1.Product Introduce

The soil oxygen sensor is applying electrochemical principles suitable for the study of monitoring soil oxygen content respiration and permeability, and the relationship between soil oxygen content and osmosis and the activity of soil microorganisms. It has the advantages of reliable performance, long service life and fast response speed. The soil oxygen sensor is a rugged and reliable oxygen concentration measurement device. Its electronic components are completely encapsulated in epoxy resin, and a filter at the front end prevents the soil particles from damaging the sensor, and can be easily buried in the soil for measurement. The measured data can reflect the change of the oxygen content in the soil in real time.

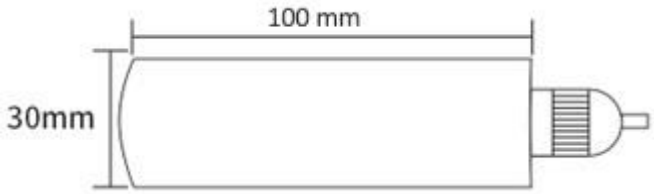
2. Product Features

1. Easy to install, can be used for long-term observation and monitoring of soil oxygen content, as well as breathing and breathability
2. All-electronic package with Nylon shell.
3. Signal output is stable.
4. High measurement accuracy, reliable performance, to ensure normal work, fast response, high data transmission efficiency.

3.Product Parameter

Measure parameters	**Value**
Measure range	0-25%VOL
Average current	≤15mA
Response time	＜15s
Output	RS485(Standard Modbus-RTU protocol)
Resolution	0.1%VOL
Accuracy	±2%FS
Power supply	7~24V
Working temperature	-20~50℃
Working humidity	15~80%(Non-condensing)
Temperature drift	0.2% FS/℃
Repeatability	≤1% FS

4. Product size

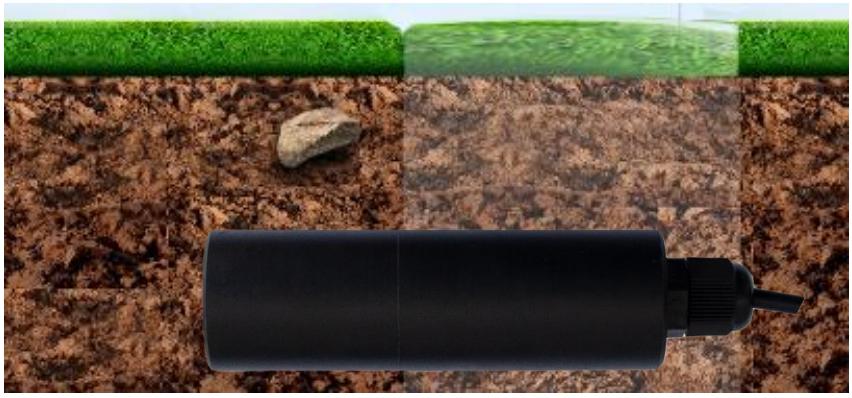


5. Measurement methods

If you are measuring on a hard surface, you should drill a hole first (the hole diameter should be bigger than the sensor probes), then insert it into the soil and compact the soil before measuring; the sensor should be protected from severe vibration and impact, let alone knocked with hard objects. Because the sensor is a black package, the sensor will heat up sharply (up to 50℃) under strong sunlight. In order to prevent excessive temperature from affecting the temperature measurement of the sensor, please pay attention to shading and protection when using it in the field or in the field.

Please note the soil oxygen sensor can be moisture-proof but cannot be water-proof, so need burry in the soil at least 10-20cm deep(according to the site condition) to avoid immersion of the sensor probe in water.

1. Buried measure method



- Make a soil profile in the vertical direction, slightly deeper than the installation depth of the bottommost sensor, between 20cm and 50cm in diameter
- Insert the sensor horizontally into the soil profile
- After the installation is completed, the excavated soil is backfilled in order, layered and compacted, and horizontal installation is guaranteed.
- If you have the conditions, you can put the removed soil in a bag and number it to keep the soil moisture unchanged, and backfill it in reverse order.

2. Multi-layers installation

6.Parsing protocol

1. Standard Modbus-RTU protocol

Baud rate: 9600bps can be set, factory default is 9600bps; Check digit: none;

Data bit: 8; Stop bit: 1

2. Read soil oxygen content at device address 0x01

- Inquiry frame

Address code	Function code	Register start address	Register length	CRC16 low	CRC 16 high
0X01	0X03	0X00 0X00	0X00 0X01	0X84	0X0A

- Response frame

Address code	Function code	Data length	Wind speed	CRC16 low	CRC 16 high
0X01	0X03	0X02	0x00 0XD1	0x78	0X18

Calculation instructions:

Soil oxygen content: The data is converted to decimal and then divided by 10

The above data indicates that the soil oxygen content is: 20.9% VOL

3. Change the soil oxygen sensor address from 01 to 02

- Inquiry frame

Address code	Function code	Reserve	New sensor address	CRC16 low	CRC 16 high
0X01	0X06	0X00 0X00	0X00 0X02	0X08	0X0B

- If success, it will feedback:

Address code	Function code	Data length	New address	CRC16 low	CRC 16 high
0X01	0X06	0X02	0x00 0X02	0x39	0X49

Note: If you forget the original address of the sensor, you can use the broadcast address 0XFE instead.

When the broadcast address 0XFE is used, the master can only connect to one slave at the same time.

7. Precautions

1. Please check whether the packaging is intact, and check whether the sensor model and specifications are consistent with the products you have purchased;
2. Do not wire with power, and only after the wire is connected and checked for error, can the power be turned on;
3. The user should not change the components and wires that have been welded before leaving the factory when the product is in use;
4. The sensor is a precision device, please do not disassemble it by yourself when using it, let alone touch the diaphragm, so as not to cause damage to the product;
5. Avoid sticky particles from entering the sensor, prevent moisture, so as not to affect the measurement performance.

8. Product warranty period

The warranty period of this product is one year. Counting from the date of shipment, within twelve months, the company is responsible for free repairs or replacements due to sensor quality problems (non-human damage), and only the cost will be charged after the warranty period.